Determining Whether Wildfire Burn Area is Affected by the Initial Ignition Source

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A. Project Highlights

This project sought to determine whether there are statistical differences in the size of wildfire burn areas between human-caused ignitions and naturally caused ignitions. Determining the difference can provide regional leaders with their local danger group and where to prioritize resource allocation to mitigate initial ignition danger or reduce fire spread.

The project focused on understanding whether the initial ignition source affects the final burn area. The project analyzed the regions where fires burned and reported the count of human ignition subtypes, such as power transmission, smoking, or firearm use. Additional factors that affect wildfire growth and damage, such as weather patterns, area fuels, suppression policy, and climate change, fell outside the scope of this project.

The analysis was built in VScode using the Python programming language, Conda Environment Manager, Jupyter Notebooks, and the following packages: sqlite3, numpy, pandas, matplotlib, ipykernal, geopandas, scipy, pptx.

Using pandas, the dataset was loaded from a SQLite database, then joined using geopandas to sort each fire into ten geographic regions used by the Interagency Fire Command Center. Then each region went through the processing pipeline steps:

1. Run statistical comparison tests for each region.
2. Create data visuals for each region.
3. Create a PowerPoint for each region using the generated visuals.
4. Report policy recommendations and conclusions for each region.

B. Project Execution

The project's goal was to determine which initial source group is more likely to cause large fires by running statistical tests at national and regional levels and delivering a report of the findings and policy recommendations to protect against large wildfires. In the project proposal, I hadn't determined how I'd choose to report my findings. Ultimately, I chose to create a slideshow for each region, breaking down the test conclusions by segment, such as standard and very large fires. I used the ADIE project planning methodology for the project:

- Analysis: Define the project goal, deliverables, scope, analysis platform, and dataset.
- Design:
    1. Perform data cleaning and prep for analysis.
    2. Run statistical tests for each region.
    3. Create visuals for each region.
    4. Report recommendations for each region.
- Implement: Write code for design steps 1-4.
- Evaluate: Review test results for each region, then create reports and policy recommendations for each region and the national level.

This framework helped break down the steps for the coding phase; I found my initial plan suitable and did not require changes at any point.

Regarding the project's timeline and milestones, I planned to complete the regional test result analysis from October 26 - 30, then complete the policy reports November 2 - 6. Ultimately, I only exceeded my anticipated timeline by one day. During development, I had solid test results and charts at the national level written in a Jupyter notebook; however, I needed an extra day to reorganize my code into a repeatable pipeline to efficiently test and report on the ten regions. After reorganizing into scripts, I was able to simplify creating the slideshow reports by using a template that handled the heavy data querying and testing at the top, then ran and visualized the segments into automated PowerPoint reports, saving hours of manual report creation.

C. Data Collection Process

While the fires dataset worked as originally planned, early in my exploratory data analysis I realized I didn't have a sufficient method to group the fires by region. States could have worked, but splitting 50 ways would overly complicate the conclusion. To overcome this, I chose to pull in the official fire regions from the National Interagency Fire Center's (NIFC) Geographic Area Coordination Centers (GACC).

While collecting data, I found little resistance. Both the fire data and the GACC areas were simple, direct zip file downloads from the United States Government. The GACC areas were a GeoJSON file. However, to use them, I had to integrate the GeoPandas package into my pipeline; from there, I essentially performed a "Left Join" between the fire data coordinates and the GACC regions to complete the dataset.

Regarding data governance issues, I used GitHub to host the repository for my analysis. The fires SQLite file was about 1 GB, which is too large for GitHub to store in a normal repository. To overcome this, I included a link to the data in my README and chose to download the file to my local environment when working across machines; however, in a

production environment, I would create a pipeline step to download the dataset directly from the USDA data repository.

C.1 Advantages and Limitations of Data Set

The USDA put great effort into compiling the datasets to ensure their cleanliness and the absence of duplicates. This was advantageous because it made it very easy to work with. The only cleaning steps I had to take were dropping a single mischaracterized fire and handling 28 fires that fell on the border of regions by assigning them to their closest match.

While acquiring the GeoJSON file was simple, using it required a specialized GIS package in Python, which meant more research, as GIS is a more specialized tool for analysis than the typical SQL and Pandas dataframes I am used to.

Additionally, while there were just over 2 million fires in the dataset, 10% had undetermined causes. The remaining fires provided a suitable sample size, but 40% of the general-cause human fires lacked known sub-causes, which may reduce the value of some human policy recommendations.

D. Data Extraction and Preparation

Both datasets were freely available as zip files from the United States Government. After unzipping, I pulled all the fire data from the SQLite table using pandas. I then pared down the query to include only the necessary columns. During EDA, I found a misclassified fire and chose to drop it, as research into the incident revealed it was later rolled into a larger complex of multiple fires. The date columns were stored in a Julian Date format; I converted them to a standard datetime in pandas, but ultimately did not use them for this analysis. Lastly, rows without a cause classification were dropped as they couldn't be used in testing. The data was otherwise very clean and did not require further processing.

The GACC regions were read in using geopandas from a GeoJSON file. Next, I performed a left join to connect the coordinates in the fires data to the region where the fire occurred. The dataset then consisted of the fire size, cause, coordinates, and region shape. These features were used to calculate the Mann-Whitney U test statistics and map the results.

E. Data Analysis Process

E.1 Data Analysis Methods

I used statistical comparison analysis to determine whether human-started wildfires are more likely to burn larger areas than those started by natural causes. This was appropriate because the fires fall into two independent groups: human-started and naturally started. Typically, this would be performed with a T-test that compares the means of the two groups; however, because fire data are not normally distributed, a specialized Mann-Whitney U test was used instead. The MWU test is thought to compare the medians of the groups instead. The test ranks the groups by a chosen ordinal value, in this case burn acreage, then compares and computes a score for which group is likely to lead to the larger value. Given the large size of the dataset, nearly every fire is likely to be statistically significant, so a rank-biserial effect size is computed to determine whether the result is both significant and meaningful.

E.2 Advantages and Limitations of Tools and Techniques

The analysis was performed using the Python programming language and useful packages such as pandas, scipy, and geopandas. This was advantageous because everything could be loaded, prepped, analyzed, and visualized in a single place using a single tool.

I did not face many disadvantages during analysis; however, while the Mann-Whitney U test did suffice for the non-normal distribution of the data, the extreme size of the data set meant that basic test results alone could not answer the project question; additional calculation of the rank-biserial effect size value was required to pair statistical significance with meaningful results.

E.3 Application of Analytical Methods

Following the exploratory and prototyping phase, the project method takes place through the following pipeline steps:

1. Load the GACC boundaries
2. Load and process fire data
    a. Query the Fire data from the fires SQLite database
    b. Load the query result into a pandas DataFrame
    c. Drop fires with unknown Ignitions
    d. Drop fires outside of Alaska or the Lower 48 United States
3. Join fires data to GACC regions
    a. Using coordinates from fires data, join fires to corresponding GACC
    b. Search for outliers that didn't land (typically on the border of a region), then approximate to nearest GACC
4. Loop through fire size segments, perform the MWU test, create result visualizations, and store them in the segment folder.
5. Loop through the segment folders, build results and charts into slide deck templates for the national level and each GACC region to meet project deliverables.

To compare the independent human vs. natural groups, I used a Mann-Whitney U test from the scipy stats library. This returned a P-value and the U statistic, a ranked value that determines whether the human group rejects the null hypothesis. Using the U statistic, I computed a rank-biserial size effect value, which estimates a regression from -1 to 1 to determine the likelihood of a fire being larger based on the initial ignition source; positive values are skewed toward human-caused fires, whereas negative values are skewed toward naturally caused fires. Results were calculated at both the national and regional levels. A region would be sorted into either the human-fires or natural-fires problem group if the result was deemed significant and meaningful. A result is significant if the alpha was below 0.005 to account for the ten regions and large number of samples, and meaningful if the |r| was greater than 0.1 to determine which group it fell into.

F Data Analysis Results

F.1 Statistical Significance

To assess statistical significance, I used the Mann-Whitney U comparison test. It returned a p-value and U statistic; I used the U statistic to calculate a rank-biserial correlation as an effect size. Given the large dataset and ten regions, I set the alpha value at 0.005. A result had to be both significant at $\alpha < 0.005$ and meaningful at $|r| > 0.1$. Following testing, it was determined that at the national level there is sufficient evidence to reject the null hypothesis and support the claim that human-caused fires have a tendency to produce larger wildfire burn areas, albeit with a somewhat weak effect on the outcome, except in Alaska.

F.2 Practical Significance

The practical significance of this project is to use the test results to make informed decisions about the direction of public policy to mitigate fire size and damage. The testing generates usable but limited results for area managers at both the national and regional level. Generally, most areas have a weak skew toward human actions creating larger fires. This does offer evidence that human activity should be targeted for reasons beyond

the fact that triple the number of human-ignited fires occur compared to naturally ignited fires, based on raw counts alone. While there may not be extreme correlation on either end, the solution provides its expected benefit when each region's report identifies which ignition source (human or natural) warrants greater scrutiny, or that neither does, and supports a clear recommendation: human-driven regions prioritize accident prevention, law enforcement, and public education; while naturally driven regions prioritize fuel management, early detection, and pre-positioned suppression resources.

F.3 Overall Success

Overall, this project is a success; creating the reporting pipeline leads to a quick, repeatable process that can rapidly generate a regional report if a leader wants to understand a different segment, or if fresh data becomes available in new years. The reporting provides adequate results to make informed decisions. Most areas with a meaningful difference fall into the human group, albeit somewhat weakly. The results aid leaders in targeting campaigns based on test results rather than just raw larger counts of 10 human fires to every 3 natural fires.

G. Conclusion

G.1 Summary of Conclusions

Wildfires are a multifaceted problem, with many factors that affect their growth and potential for damage. While understanding the effects of the initial cause is interesting and can help to guide policy, overall it is only a minor predictor of larger burn areas. The initial ignition as a single dimension is usable in a regional approach but, at scale, is likely insufficient. However, future pairings with weather data, regional fuel composition, or suppression strategies could provide an enhanced analysis and are worth future exploration. Overall, these findings are a solid stepping stone into fire analysis and have been a worthwhile development opportunity. Additionally, learning to adapt to a quickly repeatable, highly adaptable approach with a custom function library provides an efficient method to augment research visualizations and report results quickly across new data segments as fresh questions emerge, proving to be a valuable testing and reporting framework to improve understanding over time.

G.2 Effective Storytelling

Each visual supports effective storytelling by either showing the viewer the spatial impacts of fires or by giving shape to certain features of the fire data. I used Matplotlib to create each visualization.

The GACC Map provides a contextual overview to help the viewer understand how the United States fire agencies divide the country into ten similar sub-regions; it is effective in breaking the viewer out of the fifty-state model typically seen.

The GACC fire map shows the user where fires cluster and, using color, distinguishes between large and small fires. This is effective at illustrating fire effects across the country and shows how relatively few megafires there are compared to small blazes.

The regional results map provides at-a-glance insight into where a region's danger lies, without having to dive into any numbers. Results could be shared widely with just the map and key to achieve full understanding.

The R bars graph is useful for understanding how high the correlation is within the region, providing the full context of the spectrum. Drawing a line where the calculation fits creates a relative dimension and understanding that goes beyond just a decimal value.

The monthly fire bars illustrate how fire danger swells and falls throughout the year. The user can see how the four seasons affect fire counts and how fire season typically peaks in the middle of summer.

The human causes breakdown chart illustrates how human actions compare by ranking them from high to low. The visual offers swift insight into which activities to target to create the greatest reduction in fires.

G.3 Recommended Courses of Action

My first recommendation is for areas that fall under the Human Cause danger group or do not fall into either group. The majority of fires are small and started by humans. The greatest number of human fires with a known sub-cause are from Debris burning that gets out of hand and arson. Area leaders should invest in law enforcement to reduce arson and in public education campaigns to instruct on how and when to burn trash safely. Restrictions should also be put in place to ban the burning of trash and yard waste during peak fire season, as indicated by weather conditions such as low humidity and high temperatures.

The second recommendation is for areas seeking to reduce the danger posed by naturally caused wildfires. In nearly all cases, a natural wildfire is

started by lightning strikes. Fire towers should be staffed and well-maintained during the fire season to spot fires before they grow too large. Leaders should monitor weather forecasts; when storms are expected, crews should be prepositioned to suppress wildfires. Lastly, public education should be directed toward those who live close to the Wildland Urban Interface (WUI) to help them design their landscaping to reduce the chances that fires spread onto personal property.

H Panopto Presentation

Link to Panopto Presentation:
https://wgu.hosted.panopto.com/Panopto/Pages/Viewer.aspx?id=a546b5e1-b9be-4907-9c94-b4ff013d2c2b

References

McClenaghan, E. (2024, March 25). *Mann-Whitney U Test: Assumptions and example*. technologynetworks.com. Retrieved October 22, 2026, from https://www.technologynetworks.com/informatics/articles/mann-whitney-u-test-assumptions-and-example-363425

Short, K. C. (2026a). Spatial wildfire occurrence data for the United States, 1992-2024 [FPA_FOD_20260615] [Dataset]. In *Forest Service Research Data Archive*. https://doi.org/10.2737/rds-2013-0009.7

Short, K. C. (2026b). Spatial wildfire occurrence data for the United States, 1992-2024 [FPA_FOD_20260615] [Dataset]. In *Forest Service Research Data Archive*. https://doi.org/10.2737/rds-2013-0009.7

Wibbenmeyer, M., & Tastet, A. (2025, March 21). *Wildfires in the United States 101: Context and Consequences*. Resources for the Future. Retrieved October 22, 2026, from https://www.rff.org/publications/explainers/wildfires-in-the-united-states-101-context-and-consequences/?gad_source

Wildland Fire Data Management Program (DMP) - National Interagency Fire Center. (2025). *National GACC Boundaries Public* [Dataset]. Wildland Fire Data Management Program (DMP). https://data-nifc.opendata.arcgis.com/datasets/nifc::national-gacc-boundaries-public/about

Appendix A

Title of Appendix

All Python code, charts, and slide decks are available in this GitHub repository: https://github.com/JJsouth/wgu_wildfire_capstone

Fire data is available here (note the SQLite database file must be renamed to “wildfire_data” to operate properly in the script): *Forest Service Research Data Archive*. https://doi.org/10.2737/rds-2013-0009.7

GACC shape data is available here: https://data-nifc.opendata.arcgis.com/datasets/nifc::national-gacc-boundaries-public/about